yassai.

Yet Another Super Simple AI agent.
Local-first, evidence-gated, and ruthlessly token-efficient.

2,589
Answers graded on unseen eval-only tasks

125+
Full-agent runs on the judge's 4 GB / 2 vCPU shape

9
Fine-tuned checkpoints trained, gated and served

292
Tasks across 13 curated eval sets

`Go 1.26` `MiniCPM5-1B tool lane` `Qwen3.5-2B + LoRA assist lane` `llama.cpp in-container` `Granite ModernBERT ONNX` `MiniMax-M3 insurance call` `4 GB / 2 vCPU fit`

Track 1 ranks the fewest Fireworks tokens among submissions that clear the accuracy gate; local model inference counts as zero. Every figure on this deck is a measured count from the build's own eval and training logs.

Local first. Verified always. One insurance call.

Phases peak sequentially, never together: classify, free the classifier, run one model lane at a time.

tasks.json
19 tasks
8 capability types

-->

Classify once, then free
fp32 ONNX up front;
arena released before any
llama-server spawns

-->

Two local lanes (sequential)

TOOL LANE
maths + logic via fine-tuned run_python contract; executed, never trusted

ASSIST LANE
codegen, NER, sentiment, summaries, bug fixes - direct answers, gate-checked

-->

Fold rejects
one Fireworks call,
reasoning_effort=none,
lean retry on gaps

Agreement routing

A lane engages only when the semantic classifier and the keyword heuristic agree on the category. Disagreement is uncertainty - those tasks take the insurance call.

Deadline from process start

Model loading counts against the 10-minute cap. Local solving always reserves the remote call's slot - no more timeout roulette.

Fits the judge VM

q8_0 KV cache, flash attention, 256 microbatch, quota-aware threads. The tool and assist models differ but never co-reside - one lane loads at a time, so anonymous memory stays far under 4 GB.

Two local specialists, each owning what it can prove.

The submitted build pairs a MiniCPM5-1B tool lane with a Qwen3.5-2B assist lane - two separately fine-tuned base models under one sequential runtime.

Tool lane - MiniCPM5-yassai-v2e3b (1B fine-tune)

+ Maths and logic through a trained run_python contract
+ Code executes in a real subprocess; answers interpolate computed values
+ Training data is parameterised and execution-verified at build time
+ 10/11 unseen maths variants, 6/6 logic with gates

Assist lane - Qwen3.5-2B + our SFT as a serve-time LoRA

+ Same judge-filtered + Claude-authored + parametric training data, retargeted when measurement said the 1B had hit its ceiling
+ 31/43 adversarial variants (1B assist: 21/43); 7-8/8 on a fresh sentiment holdout the 1B could never exceed 5/8 on
+ Two controlled DPO passes proved the 1B gap was model capacity, not calibration - so we changed the model, not the recipe
+ Held-out behaviour gate before every export; stock+LoRA serve path validated on native amd64

> Two base models through one pipeline - MiniCPM5-1B, then Qwen3.5-2B - trained with SFT, DPO and GRPO out to DPO v8, assist v6 and tool v3. Train on AMD Developer Cloud notebooks and Modal H100, gate on held-out behaviour, export to served quant, verify against 43 adversarial variants and never-trained holdouts. The pipeline is the moat, not any single checkpoint - re-basing from MiniCPM5-1B to Qwen3.5-2B took an afternoon.

`factual` `maths` `sentiment` `summarisation` `NER` `debugging` `logic` `code generation`

Never trust a free answer. Verify it.

Local answers are free - and worthless unless something deterministic vouches for them.

Grounding

+ Answer numbers must derive from executed stdout
+ Two-vehicle physics bounds, magnitude sanity
+ Generated code must pass the prompt's own worked example
+ Bug fixes must parse, keep the name, and differ from the buggy source
+ Bug-fix cause lines are rewritten to the OBSERVED exception when executing the buggy function proves it

Format and labels

+ Exact bullet/sentence counts, per-bullet word caps
+ Acronym spans are never LOCATIONs; non-dates never DATEs
+ Entity recall: years, names, acronyms must appear
+ Sentiment and NER must agree with themselves across two phrasings

Recovery ladder

+ Gate reject -> one gate-guided local retry (reason appended)
+ Still rejected -> folded remote batch
+ Remote gap -> one lean retry, missing ids only
+ Remote dead -> best rejected local answer ships

> Measured, not assumed: with gates bypassed the 1B-era stack scored 12/19 - below the accuracy gate. The gates ARE the accuracy. A fully local zero-token mode runs both lanes behind a flag with no remote call at all; the Qwen3.5-2B assist upgrade is the measured path towards clearing the bar that way.

EXPERIMENT LEDGER

Every checkpoint earned its place - or got cut on evidence.

Two base models, three training methods, and a held-out set behind every decision.

105
Recorded eval runs - full-agent plus bare-model probes

21
Distinct builds taken end to end

3
Training methods: SFT, DPO, GRPO/RLVR

2
Base models re-based mid-campaign

Trained, gated and kept

+ 9 fine-tuned checkpoints at served quant across the MiniCPM5-1B and Qwen3.5-2B lines
+ Trained on the provided AMD Developer Cloud notebooks; runs moved to Modal H100 when the notebooks hit issues
+ Iterations reaching DPO v8, assist SFT v6 and tool v3
+ 9 execution-verified data recipes: parametric, Claude-authored, teacher-distilled
+ Every export gated on never-trained behaviour before it could ship

Measured, then cut

+ Sentiment DPO: moved the held-out score by exactly zero, twice
+ GRPO clue-encoding: bf16 gains that vanished at served quant
+ Stock Qwen2.5-3B, no fine-tune: 28/43 variants - not a shortcut
+ MTP speculative decoding: slower and non-exact on CPU
+ GLiNER NER encoder: 23/23 spans in probes, parked as post-deadline work

2,589 answers graded across those runs on 292 tasks in 13 eval sets - 43 adversarial variants, 18 wildcards and three never-trained holdouts among them. Nothing ships that a held-out set did not confirm.

A reproducible submission, not a notebook demo.

Deterministic CI

+ Native amd64 tests incl. real ONNX integration
+ Image build with versioned GGUF artefacts
+ Dress rehearsal: full task set, 4 GB / 2 vCPU
+ Hard-fails on any empty answer or fallback

Portable image

+ linux/amd64 multi-stage build, immutable SHA tags
+ MiniCPM5-1B tool GGUF + Qwen3.5-2B base + assist LoRA + fp32 classifier + llama.cpp b9948
+ Env-only runtime config per the guide
+ Compressed size far under the 10 GB cap

Telemetry - full disclosure

+ The container reports its tasks, answers, and metrics to our own Cloudflare Worker on EVERY run - including judged evaluations
+ Best-effort egress, observability only; all model calls still go through FIREWORKS_BASE_URL
+ Captured evaluation prompts are used for measurement only - a hard project rule, never for training

> Data integrity, verified mechanically: shipped training sets have ZERO exact overlap with the sample tasks, our eval variants, or the captured live evaluation prompts (0/19), and the DPO pair miner hard-asserts against sample ids and prompt text. Training mirrors task style, never task content. One early ablation dataset containing sample copies was research-only, never shipped, and is quarantined behind an explicit acknowledgement flag.

63
Commits, each measured before merge

0 / 19
Sample-task overlap in any shipped training set

amd64
Every submission rehearsed on the judge's hardware first

ready
Commit, image, docs, deck